Parking Problems

Spring 2019 ARML Power Contest

Imagine a one-way street with n parking spaces numbered $1, 2, \ldots, n$. A car turns on to the street and may park at any available space, but once the car drives past a space it cannot return and park there later.

Now imagine that each driver has a preferred space. He or she turns onto the street and proceeds to the preferred space, ignoring whether the lower-numbered spaces are empty or not. If the preferred space is available, the driver parks. If not, the driver pulls ahead and parks in the first available empty space with a higher number than their preferred space. If there are no spaces left, they cannot park their car.

Example: For a street with three parking spaces, three drivers come along with preferences 2, 3, and 2 respectively. The first driver parks in space 2, the second in space 3, but the third driver passes by space 1, then finds spaces 2 and 3 full, so they cannot park. On the other hand, even if all three drivers prefer space 1, they can all park because the first driver gets space 1, the second finds it full and proceeds to park in space 2. The third driver must go on to space 3, but at least they get to park!

A *preference sequence* of length n is an ordered list $(p_1, p_2, \ldots, p_n)$ of n positive integers between 1 and n inclusive. A preference sequence is a *parking sequence* if driver 1 has parking space preference p_1, driver 2 has preference p_2, and so forth, and every driver can park. Thus $(1, 1, 1)$ is a parking sequence, while $(2, 3, 2)$ is not.

Problem 1.

[1] (a) List all preference sequences of length 2.

[1] (b) Which of the preferences sequences of length 2 are parking sequences?

[3] **Problem 2.** For each preference sequence below, list the spaces that will be taken by each driver, in order. If the driver can't park, use an "X." For example, the sequence $(2, 3, 2)$ should have answer 23X, while $(1, 1, 1)$ will be answered 123. A more complicated example: $(2, 4, 4, 2)$ leads to 24X3.

(a) $(4, 5, 6, 3, 3, 3)$

(b) $(2, 4, 2, 4, 2, 4)$

(c) $(3, 3, 2, 2, 1, 1)$

[2] **Problem 3.** Compute, with explanation, the number of preference sequences of length n.

[2] **Problem 4.** There are 16 parking sequences of length 3. List them all.

Our ultimate goal will be to compute the number of parking sequences of length n.

[3] **Problem 5.** Prove that if any driver allows the person behind them in line to go in front of them, it does not affect whether or not the drivers can all park. That is, show that

$$(p_1, \ldots, p_{j-1}, p_j, p_{j+1}, p_{j+2}, \ldots, p_n)$$

is a parking sequence if and only if

$$(p_1, \ldots, p_{j-1}, p_{j+1}, p_j, p_{j+2}, \ldots, p_n)$$

is a parking sequence.

Applying the result of the previous problem repeatedly shows that the order of drivers actually makes no difference. That is, given any preference sequence $(p_1, p_2, \ldots, p_n)$, we can rearrange it at will without changing whether or not it is a parking sequence. For instance, to make parking more orderly we might *sort* a sequence into non-decreasing order: $(2, 4, 2, 4, 2, 4)$ would become $(2, 2, 2, 4, 4, 4)$ for example, while $(3, 3, 2, 2, 1, 1)$ would become $(1, 1, 2, 2, 3, 3)$.

[2] **Problem 6.** Prove that if preference sequence $S = (p_1, p_2, \ldots, p_n)$ is a parking sequence, then if some driver changes their preference to a *lower*-numbered space, the resulting preference sequence is also a parking sequence. That is, if any p_j is replaced by a smaller number $p_j - a$ where $1 \leq a \leq p_j - 1$, then the resulting sequence is again a parking sequence.

[2] **Problem 7.** Prove that a preference sequence is a parking sequence if and only if, when sorted, the j^{th} entry is less than or equal to j. For example, when we sort $(2, 4, 2, 4, 2, 4)$ into $(2, 2, 2, 4, 4, 4)$ the very first entry is larger than 1, so neither of these two preference sequences is a parking sequence.

The Circular Parking Garage

Instead of our one-way street, let's suppose that n cars enter a parking garage that has $n+1$ spaces in it. The car enters at space 1, and proceeds toward higher-numbered spaces until it gets to its preferred space. It then takes the first available space after that. Fortunately, this garage is circular, so that if all the spaces from its preferred space up to space $n + 1$ are all already taken, the car gets to start over at space 1. For example, if $n = 3$ and the preference sequence is $(3, 3, 3)$, then the first car parks in space 3, the second in space 4, and the third goes around the circle and parks in space 1.

[1] **Problem 8.** For the $n = 6$ preference list $(2, 4, 2, 4, 2, 4)$, parking in a garage with seven parking spaces, list the order in which the spaces are filled. (For example, in the case of $(3, 3, 3)$ above, the spaces were filled in the order 3, 4, 1.)

[2] **Problem 9.** In the example of $(3, 3, 3)$, space 2 remained empty; because there are n cars and $n + 1$ spaces, there will always be exactly one empty space. Let $n = 5$ cars park in a garage with six spaces. For each parking space k from $k = 1$ to $k = 4$, give a preference sequence for which that space remains vacant. (Your answer should be four preference sequences, one for each k from 1 to 4.)

[2] **Problem 10.** Explain why, with n cars and $n+1$ spaces in the garage, space n will always be filled. (This explains why $k = 5$ was not asked for in the previous problem!)

Since there are now $n+1$ spaces available, we might as well allow drivers to prefer to park in that last space. A *garage preference sequence* of length n is an ordered list $(p_1, p_2, \ldots, p_n)$ of n positive integers between 1 and $n+1$ inclusive. Since the garage is circular, every driver can park, and there will always be one empty space.

[2] **Problem 11.** If space k remains empty, explain why at least one driver's preferred space is $k + 1$. (Since the garage is circular with $n + 1$ spaces, we must also take our numbers modulo $n+1$, so that if space $k = n+1$ is the empty space, then $k+1$ will refer to space 1.)

[2] **Problem 12.** Let garage parking sequence $S = (p_1, p_2, \ldots, p_n)$ leave space k empty. Show that if every driver increases their preference by the same amount, the empty space is affected the same way. That is, $(p_1 + a, p_2 + a, \ldots, p_n + a)$ will lead to space $k + a$ remaining empty. (Again, numbers are to be taken modulo $n + 1$.)

Problem 13.

[1] (a) If a garage preference sequence leaves space $n + 1$ empty, explain why it must be a (regular) preference sequence.

[1] (b) If a garage preference sequence leaves space $n + 1$ empty, explain why it must, in fact, be a parking sequence.

[3] **Problem 14.** Prove that there are exactly $(n + 1)^{n-1}$ parking sequences of length n.

[2] **Problem 15.** Use the result of the previous problem to prove that $(n + 1)^{n-1} > n!$ for all integers $n \geq 2$.

Parallel Parking

Often when parking a car along the side of a street, the driver pulls just past the parking space and then backs up into the space. We will try something similar. Given a preference sequence, each driver drives to their preferred spot. If they can park there, they do so. If that spot is already taken, but the previous spot is not, they back into that space (we are no longer in the circular garage, so there is no space previous to space 1 this time). Otherwise, they drive ahead to the first empty space if there is one. For example, given the preference sequence $(2, 4, 2, 4, 2, 4)$ the first driver parks in space 2 and the second in space 4. The third driver can't park in their preferred space, so they back into space 1, and the fourth driver backs into space 3. The fifth driver can't park in space 2 or back into space 1, so they proceed to the first empty space, which is space 5. The last driver will get space 6. We will call a preference sequence a *parallel parking sequence* if all n cars can find parking spaces using the parallel parking rule. Note that $(2, 4, 2, 4, 2, 4)$ is a parallel parking sequence even though it is not a parking sequence.

[1] **Problem 16.** Of the $3^3 = 27$ preference sequences of length 3, we know there are $4^2 = 16$ parking sequences. It is a fact that there are 24 parallel parking sequences of length 3. Find the three preference sequences of length 3 that are not parallel parking sequences. (For the record, all four preference sequences of length 2 are parallel parking sequences.)

[1] **Problem 17.** Show that, as opposed to parking sequences, the order of drivers *does* matter with parallel parking. That is, show that there is a preference sequence that can be reordered into a sequence that is a parallel parking sequence, and also into an order which is not a parallel parking sequence.

[2] **Problem 18.** Show, however, that if $S = (p_1, p_2, \ldots, p_n)$ is a parallel parking sequence and $p_j \leq p_{j+1}$, then the sequence obtained from S by swapping p_j and p_{j+1} is also a parallel parking sequence.

[2] **Problem 19.** Show that every parking sequence is a parallel parking sequence.

[2] **Problem 20.** Compute, with explanation, the number of parallel parking sequences of length 4. Please try to be logical about this—a complete listing will not earn full credit.

The author does not know of a formula (either explicit or recursive) for the number of parallel parking sequences of length n. The sequence is not known to the *Online Encyclopedia of Integer Sequences*, so any information on this is probably new mathematical ground. If you have any ideas, the author would love to hear from you! Contact `power@arml2.com`.